Name:

Randa Jim

Date:

November 18th, 2023

Instructor:

Samuel Chukwuemeka

Project:

Number Theory Application: Check Digits for International Standard Book Numbers (ISBNs)

Book Title:

McGraw-Hill Education Math Grade 3

Copyright Year, Edition, and ISBNs:

© 2018

Second Edition

ISBN-10:1260019780

ISBN-13:978-1260019780

Objectives:

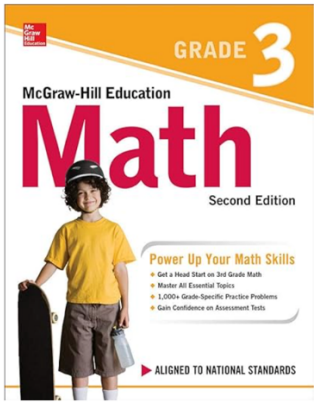
(1.) Verify the check digit for ISBN-10

(2.) Verify the check digit for ISBN-13

(3.) Convert ISBN-10 to ISBN-13

(4.) Convert ISBN-13 to ISBN-10

Calculations:



Roll over image to zoom in

McGraw-Hill Education Math Grade 3, Second Edition

Edition

by McGraw Hill (Author)

4.5 ★★★★☆ 238 ratings

Part of: Mcgraw-hill Education (18 books) See all fo

Kindle $7.99 | **Paperback $8.40 - $10.00**

Read with our free app

30 Used from $3.94
31 New from $10.00

All the Math Your 3rd Grader Needs to Succeed

This book will help your elementary school student develop the math skills needed to succeed in the classr standardized tests. The user-friendly, full-color pages are filled to the brim with engaging activities for max value. The book includes easy-to-follow instructions, helpful examples, and tons of practice problems to he master each concept, sharpen their problem-solving skills, and build confidence.

Read more

Report incorrect product information.

ISBN-10	ISBN-13	Edition	Publisher	Publication
1260019780	978-1260019780	2nd	McGraw Hill	November 3,

1st Objective: Verify the check digit for ISBN-10 (Modulo 11).

Given: 1260019780

To Calculate: the check digit

Approach: Multiplication-Modulo Approach

$1 \times 10 = 10$

$2 \times 9 = 18$

$6 \times 8 = 48$

$0 \times 7 = 0$

$0 \times 6 = 0$

$1 \times 5 = 5$

$9 \times 4 = 36$

$7 \times 3 = 21$

$8 \times 2 = 16$

Sum of the Products

$10 + 18 + 48 + 0 + 0 + 5 + 36 + 21 + 16 = 154$

Sum Divided by Modulo

$154 \div 11 = 14 \;\; Remainder\ 0$

The correct check digit is 0 and the ISBN is valid.

2nd Objective: Verify the check digit for ISBN-13 (Modulo 10).

Given: 978-1260019780

To Calculate: the check digit

Approach: Multiplication-Modulo Approach

$9 \times 1 = 9$

$7 \times 3 = 21$

$8 \times 1 = 8$

$1 \times 3 = 3$

$2 \times 1 = 2$

$6 \times 3 = 18$

$0 \times 1 = 0$

$0 \times 3 = 0$

$1 \times 1 = 1$

$9 \times 3 = 27$

$7 \times 1 = 7$

$8 \times 3 = 24$

Sum of the Products

$9 + 21 + 8 + 3 + 2 + 18 + 0 + 0 + 1 + 27 + 7 + 24 = 120$

Sum divided by Modulo

$120 \div 10 = 12 \;\; Remainder\ 0$

The correct check digit is 0 and the ISBN is valid.

3rd Objective: Convert ISBN-10 to ISBN-13

Given:1260019780

To Convert: to ISBN-13

1. Remove check digit: 126001978
2. Append 978: 978126001978
3. Calculate the check digit: I already calculated it to be 0.

ISBN-13: 978-1260019780

4th Objective: Convert ISBN-13 to ISBN-10

Given: 978-1260019780

To Convert: to ISBN-10

1. Remove the check digit: 978-126001978
2. Remove the 978: 126001978
3. Calculate the check digit: I already calculated it to be 0.

ISBN-10: 1260019780

References (APA7):

Chukwuemeka, S. D. (2023). *Number Theory Application Check Digits for International Standard Book Number (ISBN).* Mathematics Education Courses. Retrieved from https://mathematicseducation.appspot.com/.

McGraw-Hill Education Math: Grade 3 (Second). (2018). . McGraw-Hill.